

Boyang 'Boe' Zhou

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Education

University of Washington, Seattle, WA M.S. in Computer Science, GPA 3.9/4.0	Anticipated 6/2026
University of Washington, Seattle, WA B.S. in Computer Science, Minor in Environmental Studies, GPA 3.96/4.0	9/2021 – 3/2025

Research experience

Graduate Student Researcher - Augmented Reality & Computer Vision Makeability Lab , University of Washington, Seattle, WA	1/2025 – Present
- Developed a LiDAR-based sensing pipeline with a privacy-preserving design for mobile devices, enabling real-time obstacle mapping to assist users who are blind or have low vision in navigation through tactile and audio feedback. - Fine-tuned a YOLO model on a curated dataset and optimized it using CoreML, achieving 60 fps real-time on-device inference to power a visual augmentation system for live soccer events.	
Student Researcher - Video Diffusion Reality Lab , University of Washington, Seattle, WA	6/2022 – 12/2024
- Developed novel joint fine-tuning pipeline for video diffusion models to achieve temporally-consistent frame interpolation, unlocking new application for video generation methods. - Curated motion-rich video dataset and orchestrated large-scale training experiments across HPC clusters to evaluate model performance under varying architectural configurations. - Built AR-based lumbar injection training simulator integrating finite element analysis for realistic spine deformation and HRI-based spine localization, enabling a prototype for radiation-free Epidural Steroid Injection procedural.	

Publications

Generative Inbetweening: Adapting Image-to-Video Models for Keyframe Interpolation <i>Xiaojuan Wang, Boyang Zhou, Brian Curless, Ira Kemelmacher, Aleksander Holynski, Steve Seitz</i>	ICLR 2025
CARE - Clinician Augmented Reality Environment: Design and Technical Feasibility of Apple Vision Pro for Image-guided Surgeries <i>Wang et al. (including Boyang Zhou)</i>	To appear in JAMIA

Work experience

Grant and Project Coordinator University of Washington Sustainability , Seattle, WA	6/2023 – 6/2025
- Directed a portfolio of over 50 campus sustainability projects (solar, biodiversity, environmental justice) funded by a \$500,000/year grant. - Served as the primary liaison between project teams and diverse stakeholders, including tribal representatives, infrastructure engineers, and research groups. - Implemented participatory budgeting and decision-making practices with an eight-member student committee, becoming the first organization at the university to establish such a governance structure.	
Software Engineer Intern - GIS Asset Management King County Department of Natural Resources & Parks , Seattle, WA	6/2024 – 12/2024
- Engineered and deployed new features for Lucity GIS, integrating disparate geospatial (ArcGIS) and asset databases. This streamlined infrastructure maintenance workflows, resulting in a significant reduction in manual data entry for park rangers. - Conducted spatial analysis in ArcGIS on county demographic data to identify underserved communities, directly influencing the trail master plan by identifying 5 high-priority areas for new trail development in previously underserved communities.	

AI Software Engineer Intern

Hugsi, Bellevue, WA

6/2023 – 9/2023

- Architected an error-aware RAG pipeline using GPT-3.5 for a Flutter-based iOS app, improving the relevance of personalized gift suggestions on homepage and reducing hallucinations by 65% based on internal evaluations.

Leadership/Teaching experience

Teaching Assistant CSE557: Computer Graphics

Paul G. Allen School of Computer Science & Engineering, Seattle, WA

9/2025 – Present

- Teach 20 masters and PhD students code their first 3D modeling, rendering, and animation software.
- Develop new functionality for CMU’s Scotty3D engine to tailor to instruction needs including improved test cases, segfault debugging, and auto-save functionalities, improved accessibility of course material compared to previous years.

Club President

Extended Reality Association at the University of Washington, Seattle, WA

11/2022 – 9/2025

- Led a team of 7 officers and over 80 student to build an interdisciplinary Extended Reality community in the Seattle area through hosting technical workshops, speaker series, and hackathons.
- Secured over \$60,000 to purchase Vision Pros and Quest headsets accessible to all students.
- Led a student project that utilized LiDAR to capture campus buildings and develop indoor accessibility maps for virtual visits and identifying accessibility issues.

Talks

"Measuring the impact of student-led sustainability"

Association for the Advancement of Sustainability in Higher Education Conference

2025

"Leading justice-centered sustainability"

Washington Oregon Higher Education Sustainability Conference

2023, 2024

Fellowships

Mary Gates Scholar

\$5,000

2024

Technical skills

Proficient Languages: Java, Python, Rust, C+ +, Typescript

Other: Swift, C#, SQL, GLSL, Docker, PyTorch, Unity, OpenCV, TensorFlow, React, OpenXR, RealityKit, Three.js, ArcGIS, QGIS